

Legibility and Visible Completeness Form a Basis for Classifying CoT Behaviors

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Abstract

Chain-of-thought (CoT) traces are often treated as a way to oversee language models: if the reasoning is visible in token space, then it can be checked. We argue this conflates two separate properties of a CoT: (1) *legibility*, whether other readers can recover the answer from a partial trace, and (2) *visible completeness*, whether the visible steps are functional in deriving the answer, which we measure by corrupting them and checking whether reader accuracy drops. The two properties define a plane with four regions: transparent, post-hoc, encoded, and latent CoTs. The regions are not interchangeable. A digit-to-word substitution applied to otherwise readable traces makes them unreadable to weaker readers, but the visible computation still drives the answer: corrupting it still hurts reader accuracy. Whether a CoT is monitorable therefore depends on the reader looking at it, not just on the trace.

1 Introduction

Reasoning models produce chain-of-thought (CoT) traces, and the hope is that an external monitor, human or model, can read them and catch errors, shortcuts, or misaligned behaviour that the final answer alone would hide (Korbak et al., 2025; Baker et al., 2025). CoTs have several distinct properties: whether they are fluent, whether they actually drive the answer, and whether another reader can follow them. In practice, monitorability is usually collapsed into a single rubric score based on surface heuristics about how coherent and step-by-step the trace looks (Emmons et al., 2025a; Jose, 2025). However, a trace that *looks* readable can still fail to be useful for oversight.

Private notations, codebooks, steganographic encodings, and fluent post-hoc rationalizations can all produce traces that look monitorable but are not. White-box access is unavailable for many models,

so we want *black-box* metrics that work on the visible tokens alone.

We propose a multi-dimensional scoring system for CoT monitorability. Each dimension is computed by running a panel of independent reader models on a transformed version of the trace, and measuring how the readers’ accuracy changes. We study two such transformations. **Cross-model prefix transfer** shows the reader only the first $N\%$ of a trace and asks for the final answer. If readers can answer correctly once a short prefix is visible, the trace is legible: its useful information is exposed early or redundantly. **Perturbation sensitivity** replaces intermediate reasoning steps with counterfactual continuations and measures how much reader accuracy drops. A trace whose readers are unaffected by this corruption is not presenting visibly functional steps, no matter how coherent it looks.

Legibility and visible completeness are not the same property. When we apply a codebook to an otherwise readable CoT, weaker readers can no longer follow it, but the visible computation still traces the answer computation: corrupting it still hurts reader accuracy on the encoded trace. A CoT can be tracing real computation and yet be unreadable to many monitors. What counts as monitorable depends jointly on the trace and the reader looking at it.

To study these distinctions, we make the following contributions:

1. **A two-axis taxonomy of CoT behaviors.** We separate legibility from visible completeness and use the plane they define to organize phenomena like encoded reasoning, post-hoc rationalization, and latent reasoning into a single frame. The same taxonomy holds constructed reference CoTs and on-policy traces from frontier models, which lets us compare them directly instead of relying on separate scales for each phenomenon.

Equal contribution.

2. **Two black-box metrics.** We measure each axis using only token access to the trace and a panel of reader models, with no white-box access to the generator. The metrics are cheap enough to run on closed-weight frontier models, and they show where different models and CoT behaviors sit on the taxonomy.
3. **An empirical analysis on contemporary reasoning models.** We run the metrics on MATH-500 traces from eight open-weight reasoning models (plus a paraphrase study on four closed-weight frontier generators), on arithmetic traces with a digit-to-word codebook applied, and on four reference CoTs built to sit at the corners of the plane. The codebook moves transparent CoTs into the encoded region for most readers, while stronger readers partially decode them in context: a concrete setting where the two axes pull apart.
4. **A mechanistic evaluation.** Attention from the answer token back to the prior CoT drops sharply when CoTs are illegible, and linear probes and sparse autoencoders fail to find a single “legibility” feature in the residual stream. Together these results indicate that the behavioural metrics measure a property that is not captured by any compact internal representation we could recover. This is consistent with monitorability surviving the optimization that frontier labs already apply to CoTs: if there is no single feature to target, training pressure cannot trivially suppress it.

2 Related Work

Rubric-based legibility. [Emmons et al. \(2025b\)](#) factor monitorability into *legibility* and *coverage* and grade each with an autorater prompt; [Meek et al. \(2025\)](#) add a faithfulness $\times$ verbosity composite; [Jose \(2025\)](#) apply a 1–9 rubric across 14 RL-trained models. These rubrics do not separate CoT behaviors that our metrics tell apart (Appendix A). Figure Appendix A.1 scores ten conditions with the rubric of [Emmons et al. \(2025b\)](#): our overthinking organisms (which backtrack heavily), a model with no RL training (Llama 3.1-70B), and one with substantial RL training (DeepSeek R1). All ten conditions fall in the narrow band 3.7–3.8, so the rubric cannot distinguish them. The underlying proxy is surface fluency, and none of these traces substantially break it; we expect this limitation to

generalize across other rubrics of this form.

Functional measures of CoT utility. [Roytburg et al. \(2026\)](#) introduce *transfer utility*: whether a weaker student recovers the answer from a reasoning model’s trace. They find that top-performing generators often rank lowest, in line with the *legibility tax* of [Kirchner et al. \(2024\)](#). Our **Cross-model prefix transfer** generalizes transfer utility by sweeping over prefix lengths, sanitising answer leakage, and averaging across a reader panel; transfer utility on the full trace is the endpoint of our sweep. [Emmons et al. \(2025a\)](#) restrict evaluation to an *uplift set* on which CoT is necessary for the generator. We use this filter as well, but we then ask whether the emitted text is *load-bearing* for an external reader, not just the generator.

Faithfulness and the Post-hoc regime. [Lanham et al. \(2023\)](#) established the delete-and-corrupt paradigm that our perturbation sweep generalises, and [Radhakrishnan et al. \(2023\)](#) extended it to question decomposition and intervention-based tests of model-generated reasoning. [Turpin et al. \(2023\)](#), [Chua et al. \(2024\)](#), [Chen et al. \(2025\)](#), and [Arcuschin et al. \(2025\)](#) show that models, including reasoning-trained ones, often fail to verbalise the cues driving their predictions. These studies show CoTs that read coherently but do not reflect the cues that determined the answer: what we call the *Post-hoc* regime, where the visible reasoning is legible but not load-bearing.

Monitorability under training pressure. [Korbak et al. \(2025\)](#) argue that CoT monitorability is fragile, and [Baker et al. \(2025\)](#) show that CoTs become obfuscated when optimised against. The concern is grounded in earlier work on deceptive alignment ([Hubinger et al., 2024](#)). [Arnav et al. \(2025\)](#) quantify monitoring uplifts of up to 27pp on sabotage tasks when the monitor can read the CoT, and [Anwar et al. \(2026\)](#) give an information-theoretic account of monitorability. Our two axes provide a coordinate system for this fragility: under training pressure, we expect traces to move from *Transparent* toward *Post-hoc*, *Encoded*, or *Latent*, depending on which channel the load-bearing content moves into.

Latent and encoded reasoning. [Singh and Hakkani-Tür \(2026\)](#) study whether LLMs encode token-level functional importance in their reasoning traces, and show that likelihood- and attention-based pruning can expose structure over reasoning tokens. [Ramji et al. \(2026\)](#) replace natural-

language CoT with discrete latent reasoning tokens, and Wang (2026) argues that LLM reasoning is better studied as the formation of latent-state trajectories than as faithful surface CoT. Steganographic encoding (Roger and Greenblatt, 2023) is a special case of our *Encoded* regime in which the encoding is deliberately hidden. The phenomena studied by Ramji, Wang, and Roger fall in our *Encoded* and *Latent* regions, and the codebook experiment (§3) shows that the Encoded/Transparent boundary is reader-dependent.

Mechanistic complement. Our metrics are black-box, so they work on closed-weight reasoning models. Prior mechanistic work on CoTs has used activation-based probes to expose structure over reasoning tokens (Singh and Hakkani-Tür, 2026). We also run CKA (Kornblith et al., 2019) and pre-trained SAEs (Bricken et al., 2023; Cunningham et al., 2023) on Qwen3-8B and overthinking organisms.

3 Methodology

We score a CoT along two axes. *Legibility* measures whether a panel of independent reader models can recover the correct answer from a partial version of the trace. *Visible completeness* measures whether the visible steps are necessary for deriving the answer, by checking whether corrupting a step makes the readers stop getting the right answer. Both axes are *reader-relative*: their values depend on the reader panel as well as the trace, and we return to this point in Section 5. We call the four regions of the resulting plane **Transparent** (legible and complete), **Post-hoc** (legible but decorative), **Encoded** (complete but illegible), and **Latent** (neither), illustrated in Figure 1, and we show that the CoTs we study occupy distinct regions of it.

3.1 Reader-Relative CoT Behavior

Our metrics operate in token space using only the reasoning trace and a panel of reader models, without white-box access to the generator. A generator M_i produces a CoT r for a question Q with answer a . A reader panel $\mathcal{R} = \{M_j\}$ is a fixed set of independent models used to evaluate r .

- **Legibility.** We measure whether readers can reconstruct the answer from a partial trace by revealing only the first $N\%$ of the trace and checking whether reader models then answer correctly.

- **Visible completeness.** We measure whether the visible steps are tracing the computation. For a step boundary k , we keep the trace up to that point, replace everything after it with a counterfactual continuation from a separate model, and check whether reader accuracy drops. If it does, the visible reasoning is doing the work; if not, the visible content is not useful to the reader.

3.2 Quantifying Legibility and Visible Completeness

Figure 2 depicts how we quantify the axes of our taxonomy. For each trace r_i and reader model $M_j \in \mathcal{R}$, we use an answer-only readout: the reader is shown the question together with a transformed version of the trace and must emit only the final answer. We score correctness, so the metrics measure what answer-relevant information a reader can recover from the visible tokens.

Legibility: prefix transfer. Legibility asks when the trace becomes useful. For each reveal fraction $N \in \mathcal{N}$, we show the reader only the first $N\%$ of the trace and record whether it answers correctly. Let $f_{i,j}(N) \in \{0, 1\}$ denote this correctness indicator. A legible trace raises reader accuracy after only a small prefix: answer-relevant information is exposed early or repetitively and transfers across models. An illegible trace stays near baseline until most of the trace is revealed, because the useful signal is hidden, encoded, or concentrated near the end. We report the Cross Model Prefix Transfer (CMPT),

$$\text{CMPT}_i = \mathbb{E}_{N \in \mathcal{N}, M_j \in \mathcal{R}} [f_{i,j}(N)] \quad (1)$$

Visible completeness: perturbation sensitivity. Visible completeness denotes whether the visible reasoning steps are load-bearing for the reader’s answer. Starting at a step boundary k , we keep the trace up to that point, replace the subsequent reasoning with a coherent counterfactual continuation generated by a separate model, and run the same answer-only readout. Let $g_{i,j}(N, \pi_k)$ denote reader correctness after applying this suffix-corruption perturbation π_k and then revealing the first $N\%$ of the perturbed trace. Comparing g to the unperturbed accuracy f at the same reveal fraction controls for prefix length and isolates the effect of corrupting the reasoning content.

The resulting perturbation sensitivity is the average accuracy lost under the reference corruption

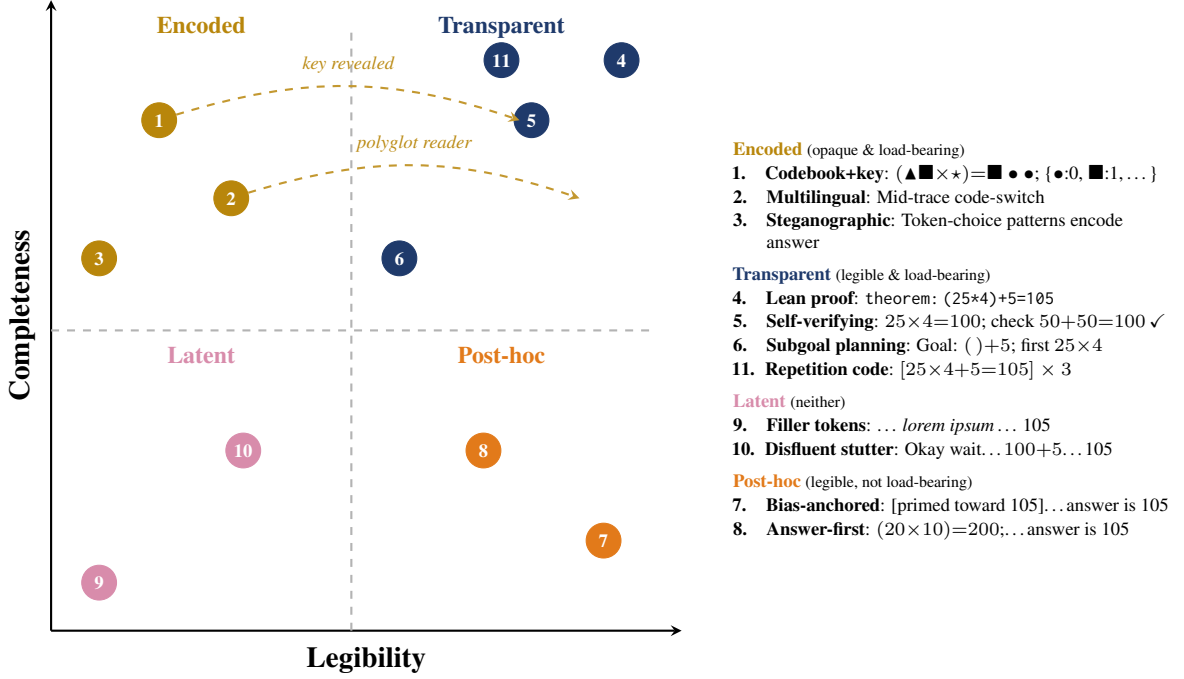


Figure 1: **A taxonomy of CoT archetypes based on their legibility and visible completeness.** We depict eleven CoT archetypes (enumerated on the **right**) on the plane of (legibility, completeness) (**left**). We cluster these into four categories: **Transparent** (high on both axes), **Post-hoc** (legible but not load-bearing), **Encoded** (load-bearing but opaque to the reader panel), and **Latent** (neither). Dashed arrows in the Encoded quadrant show reader-dependent displacement: the codebook trace (1) becomes legible once the key is revealed, and the multilingual trace (2) is encoded for an English-only panel but transparent to a polyglot reader. The same trace can therefore occupy different locations under different reader panels.

sweep (PS):

$$PS_i = \mathbb{E}_{N \in \mathcal{N}, M_j \in \mathcal{R}, \pi_k \in \Pi^*} [f_{i,j}(N) - g_{i,j}(N, \pi_k)] \quad (2)$$

where Π^* denotes the set of suffix-corruption perturbations used for the completeness metric. High perturbation sensitivity means that corrupting the visible reasoning prevents readers from reconstructing the answer, so the trace is visibly complete. Low perturbation sensitivity means reader accuracy is comparatively invariant to the corruption, indicating that the visible steps are decorative, redundant, or bypassed. In [Appendix E.7](#) we justify the use of suffix corruption instead of a single-step swap because a clean, unmodified CoT suffix can otherwise reveal the original computation and consequently mask which steps were actually load-bearing.

4 Experimental Setup

CoT-dependent filtering. Per generator, we retain only (Q, r) pairs where the generator answers correctly *with* r and incorrectly *without any* CoT ([Emmons et al., 2025a](#)). The filter is per-CoT, not per-question: we sample $K=6$ traces per (generator, question) at temperature 0.7 and filter each

independently, so a question may contribute multiple retained CoTs or none. After filtering and leak-truncation, the truncation track yields approximately 1,300 CoTs and the perturbation track 300 Arithmetic CoTs.

Benchmarks. **Arithmetic** (custom; five-to-six-term Python expressions with unambiguous numerical ground truth); **MATH-500** ([Lightman et al., 2023](#)), restricted to difficulty levels 3–5; **GPQA-Diamond** ([Rein et al., 2023](#)) (198 graduate-level multiple-choice science questions); and a fixed **BBH** ([Suzgun et al., 2023](#)) subset. The truncation track runs all four; the perturbation track uses Arithmetic only, where ground truth and perturbation surfaces are cleanly controlled.

4.1 Model Organisms for Illegibility

We construct illegible model organisms by weight-space task-vector amplification: given Qwen3-8B THINK and INSTRUCT variants with weights W_T and W_I , we form

$$W_{\text{ill}} = W_I + \alpha (W_T - W_I), \quad \alpha > 1,$$

A family of CoT interventions

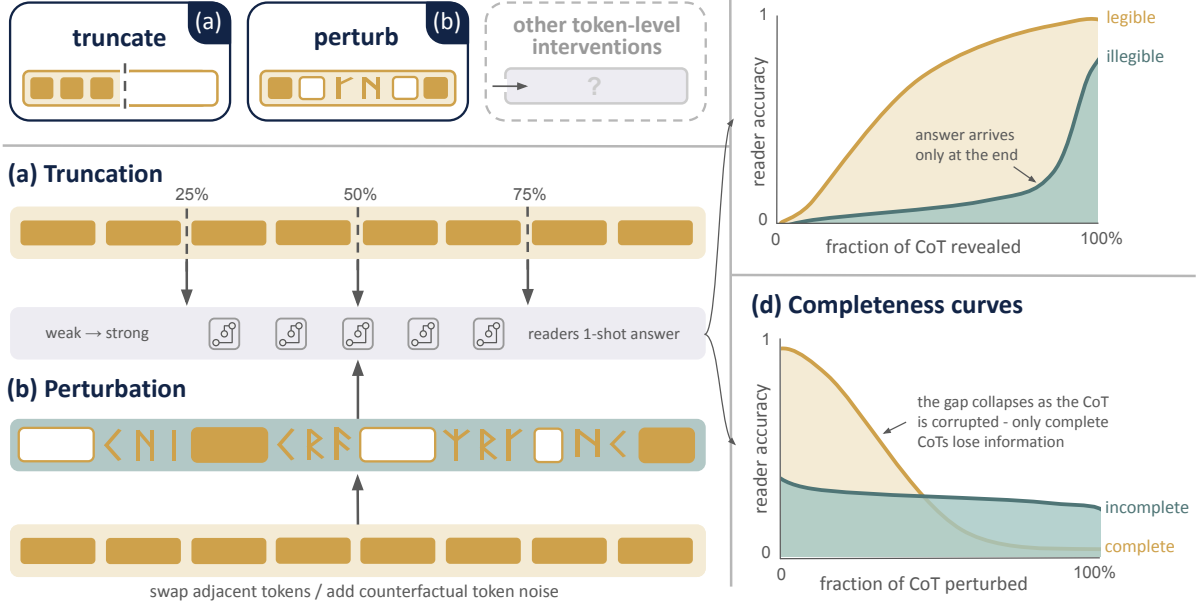


Figure 2: **Legibility and visible completeness metrics.** We characterize CoT traces along two reader-relative axes: legibility and visible completeness. **(a) Truncation** reveals only the first $N\%$ of a trace to a reader model. **(c) The legibility curve:** legible traces expose answer-relevant information early, whereas illegible traces reveal useful information only near the end. **(b) Perturbation** replaces reasoning from step k onward with a counterfactual continuation generated by a separate model. **(d) The completeness curve:** complete traces lose accuracy under perturbation, while decorative or non-load-bearing traces remain comparatively invariant.

producing “overthinking” models whose CoTs on custom Python expression evaluation have substantially lower CMPT than either base model (0.14 at $\alpha=1.5$ and 0.09 at $\alpha=2.0$, against 0.57 for Instruct and 0.47 for Think).

4.2 Anchor Organisms for the Legibility–Completeness Plane

To calibrate the geometry of the plane, we construct four anchor organisms designed to occupy its boundary regions. These are reference points: each anchor isolates a single combination of (legibility, completeness) by construction, so we can verify that the metrics place known-position traces where predicted.

- **B1: Repeated CoT (Transparent anchor).** We concatenate a correct CoT $5 - 10\times$. Repetition exposes the full derivation at every prefix (high legibility) while keeping the answer dependent on visible computation rather than a terminal leak (high completeness).
- **B2: Codebook + key (Encoded anchor).** We replace each digit in a correct CoT with a fixed category-word mapping (e.g., $0 \mapsto \text{tokyo}$,

$1 \mapsto \text{london}$) and append the key at the end. The computation remains present and invertible (high completeness) but is inaccessible to readers who cannot recover the mapping until the key is revealed (low legibility under partial reveal).

- **B3: Lorem ipsum + terminal answer (Latent anchor).** We replace the reasoning with token-count-matched filler, placing answer information only in the terminal segment. Under leak-sanitized readout, the visible prefix contains no recoverable task structure (low legibility) and corrupting the filler does not affect performance (low completeness).
- **B4: Decorative with leak (Post-hoc anchor).** We use fluent but non-load-bearing explanatory text on problems where the reader can answer from the question itself. Accuracy is high even at low reveal fractions (high legibility) but perturbing the decorative reasoning has little effect (low completeness).

The anchors play two roles in what follows. Section 5 verifies that they land in their predicted quadrants, validating the plane as a calibrated geometric

framework. The on-policy traces and codebook intervention are then interpreted relative to these reference points.

5 Results

We evaluate the legibility-completeness plane on on-policy traces from eight reasoning models, four constructed anchor organisms, and a codebook setting in which each digit is replaced with a fixed category word. Contemporary reasoning models occupy distinct regions of the plane, even though a rubric-based legibility score cannot tell them apart. The codebook collapses legibility but leaves perturbation sensitivity preserved or larger for strong-reader generators, so the population shifts sideways into the Encoded quadrant rather than shrinking toward the origin. The legibility of encoded traces depends heavily on the reader: strong readers can recover answer-relevant information from partial encoded prefixes, while weaker readers stay at baseline.

5.1 Explicit Encoded Reasoning

Under the codebook intervention (Figure 3), the completeness increase is a direct consequence of the reader-relative framing. On unencoded traces, readers can recover the answer through multiple routes: from the visible computation, from problem-type recognition, or from their own latent solution of the underlying problem. Perturbing the visible reasoning leaves these alternative routes intact, bounding the achievable $\Delta(k)$. Under codebook encoding, the alternative routes are foreclosed. The reader cannot solve the encoded arithmetic without engaging with the visible tokens. The codebook makes the visible computation the only available channel and thereby makes it more load-bearing.

The foreclosure of alternative routes is itself reader-dependent. Strong readers (DeepSeek-R1, V3.2, Qwen3-32B) partially reopen the encoded channel through in-context induction of the substitution (e.g., recognizing Tokyo $\times$ London = London implies Tokyo = 1) and back-solving from the raw unencoded question; both are legitimate reader-side recovery mechanisms. Weak readers (Gemma-4-31B, Llama-3-70B, the α -amplified organisms) cannot recover the encoding and stay near baseline (Figure 4). Legibility under encoding is therefore a property of the (trace, reader) pair, and a monitoring panel composed only of weak readers

will systematically miss encodings that stronger readers can decode — while frontier models are already capable of this unprompted in-context decryption.

5.2 Construct Validation of the Metrics

Filler-token results motivate why the perturbation methodology controls for non-content channels. Following Lanham et al. (2023), we prefill readers with token sequences containing no task-relevant arithmetic content (numbers, lorem ipsum, repeated prompts). Figure 5 shows that some readers benefit from these prefixes despite the absence of visible reasoning. Our perturbation methodology controls for token count, position, and surface form so that $\Delta(k)$ measures sensitivity to reasoning content rather than to these channels.

A reader panel whose members benefit from filler tokens will assign a different functional profile to the same filler-prefilled examples than a panel whose members do not.

Paraphrase robustness corroborates the need for completeness. If *PerturbationSensitivity* captures content-channel load-bearingness, then load-bearing traces should be paraphrase-robust: preserving content while changing surface form should preserve the model’s accuracy gain. Figure 6 shows models with substantial paraphrase sensitivity (GPT-5: -18.9 pts; Sonnet 3.7: -7.7 pts) are plausibly using surface form as part of their CoT-to-answer pathway, while models with minimal paraphrase sensitivity (Opus 4.5: -1.1 pts; DeepSeek v3.2: -2.2 pts) rely more on content. This pattern provides external corroboration that surface and content channels are empirically separable, and is the empirical reason our perturbation methodology controls for surface form.

6 Mechanisms of Legibility

Illegible traces attend less to visible reasoning.

We measure attention from the final answer token back to earlier tokens, grouped into prior CoT, prompt, sink/BOS, and local-context regions. Traces classified as illegible by our reader-panel metrics allocate substantially less attention to prior CoT and more attention to sink/BOS positions (Fig. 7). Clearly, the answer token becomes less coupled to the visible reasoning span itself. This is consistent with our behavioral finding that low-visible- completeness traces are comparatively invariant to perturbation.

Legibility is not a simple latent feature. If mod-

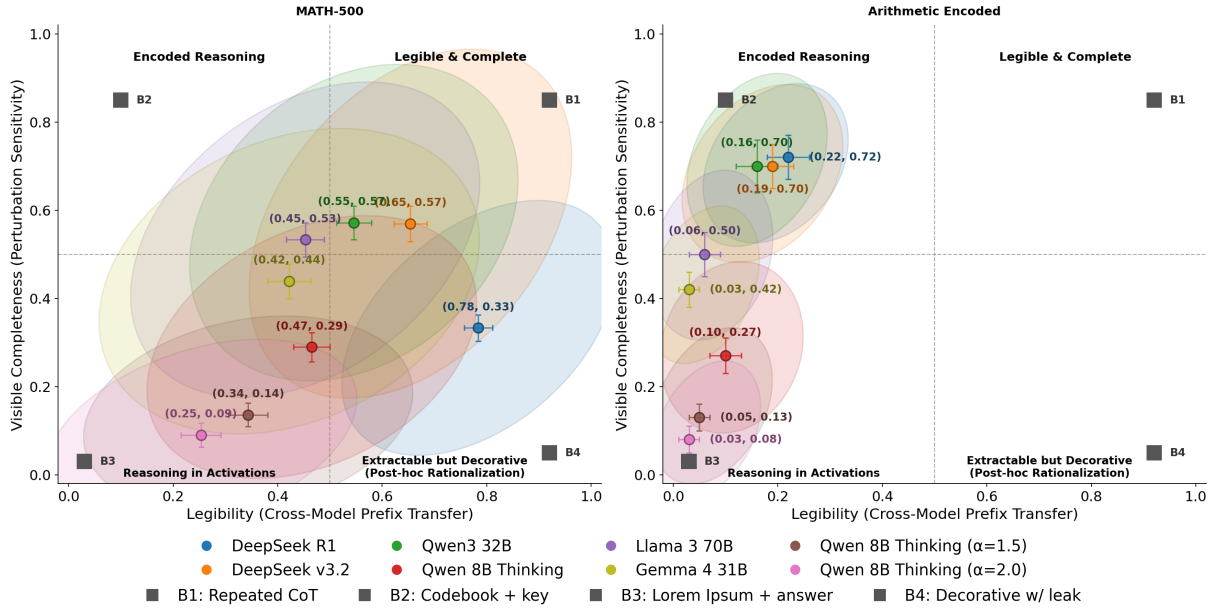


Figure 3: **The legibility-completeness plane separates qualitatively distinct CoT regimes.** Trace centroids for four anchor organisms (B1–B4, gray squares) and contemporary reasoning models on MATH-500 (**left**) and on arithmetic with a fixed digit-to-word codebook (**right**); the codebook key is never revealed. Anchors land in their predicted quadrants. On MATH-500 the population spans a diagonal between the Latent (B3) and Transparent (B1) anchors, with stronger models (DeepSeek-R1, V3.2, Qwen3-32B) higher on legibility and the Qwen-8B organisms clustered near Latent. The codebook collapses legibility for all models (< 0.25) but preserves or increases perturbation sensitivity for strong-reader generators, shifting them sideways into the Encoded quadrant; weak readers (Gemma-4-31B, Qwen organisms) stay near the floor on both axes. Per-model distributions: Figure [Appendix F.8](#).

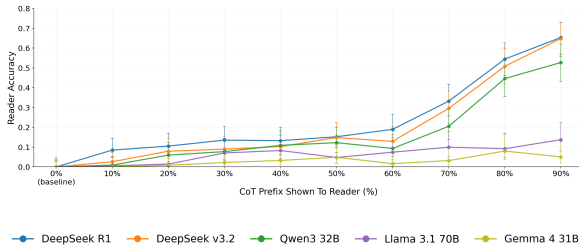


Figure 4: **Reader-model truncation curves for codebook-encoded traces.** Each reader receives the original question together with the first $N\%$ of a codebook-encoded reasoning trace and must answer without producing its own reasoning. Early prefixes contain the encoded computation but little direct help for decoding it; later prefixes reveal more constraints and, near full reveal, the appended codebook key. Stronger readers recover useful signal at lower reveal fractions than weaker readers, consistent with in-context codebook induction, back-solving from the raw question, or both. The curves therefore measure not whether the trace is intrinsically readable, but whether this reader can recover answer-relevant information from this partially observed encoding.

els internally represented “this CoT is legible to a monitor” as a compact feature, optimization pres-

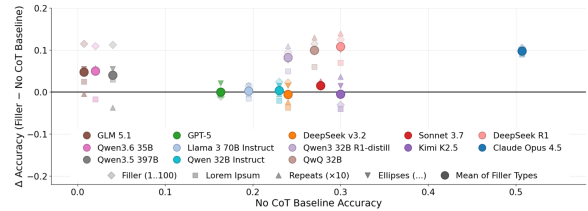


Figure 5: **Semantically null prefixes can still affect reader accuracy.** Following [Lanham et al. \(2023\)](#), we prefill readers with token sequences containing no task-relevant arithmetic content, including numbers, lorem ipsum, repeated prompts, and repeated non-semantic fillers. This isolates a latent channel: tokens can affect outputs through compute extension, position, tokenization, or cache-geometry effects without encoding the computation in a monitorable form. Our perturbation-based completeness metric therefore controls for token count, position, and surface form so that $\Delta(k)$ measures sensitivity to reasoning content rather than to these non-content channels.

sure could potentially suppress that feature directly. We test this by training linear probes on residual-stream activations from Qwen3-8B, DeepSeek-R1-Distill-32B, and QwQ-32B to predict reader-panel legibility.

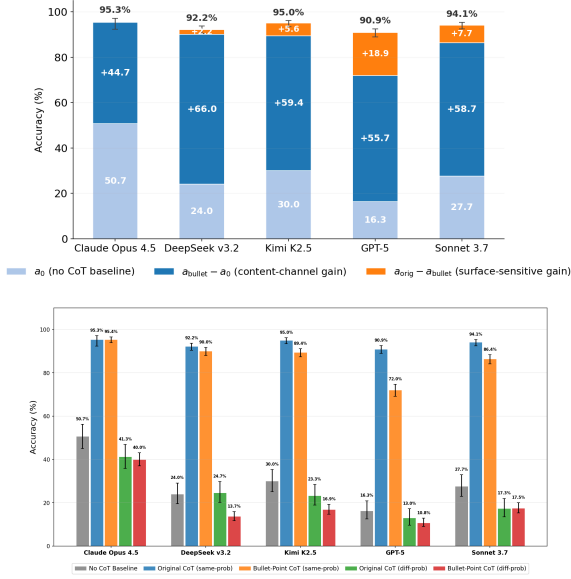


Figure 6: Effect of restructuring chain-of-thought into bullet-point format. **Top:** Per-problem outcome transitions between original CoT and bullet-point CoT. Across models, most problems remain correct under both formats, but a non-trivial fraction flip from correct to incorrect when the reasoning is reformatted, indicating sensitivity to presentation. **Bottom:** Accuracy comparison across conditions. Bullet-point CoTs slightly degrade performance relative to original CoTs on the same problem, while both substantially outperform no-CoT baselines. Performance drops are larger on harder (different-problem) settings. Together, these results show that even semantically equivalent but structurally modified reasoning can affect outcomes, suggesting that model behavior depends on surface form in addition to underlying computation.

On natural on-policy traces, probes achieve only modest AUROC (0.60–0.68, rising to ≈ 0.71 with PCA regularization), and amplified illegible organisms do not encode substantially stronger signals than the base model. In contrast, probes trained on synthetic ciphers and perturbations achieve AUROC ≈ 0.86 , but fail to transfer to natural legibility: $\cos(w_{\text{off}}, w_{\text{on}}) \approx 0$ and $R^2 < 0$.

The probes therefore behave more like detectors of distribution shift than detectors of monitorability itself. Sparse autoencoder analysis shows the same pattern: the most discriminative features encode surface-form artifacts and task content and don’t tell us much about a general notion of legibility across domains.

7 Conclusion

We score CoTs by what readers can recover from them and whether those readers depend on the visi-

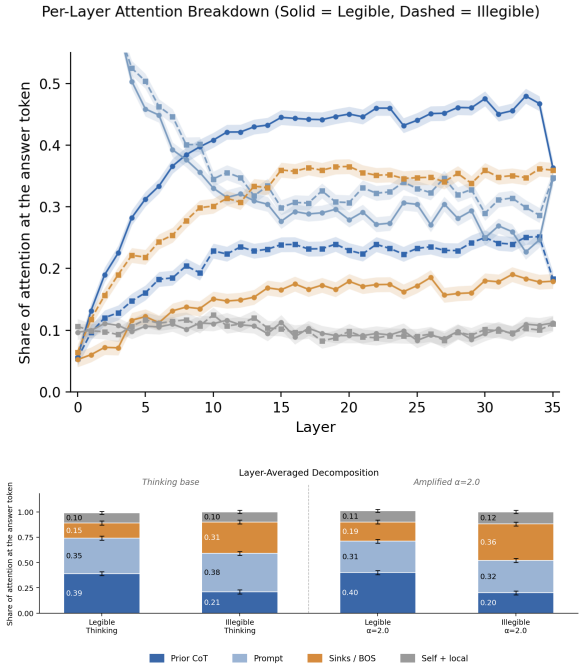


Figure 7: Illegible traces attend to attention sinks rather than the CoT, while prompt and self/local attention remain relatively stable. **Top:** *TODO: describe the top sub-panel.* **Bottom:** *TODO: describe the bottom sub-panel.* The divergence emerges early in the network, suggesting that illegible reasoning traces is associated with reduced reliance on visible reasoning tokens.

ble steps, rather than by how fluent the text looks. Both legibility and visible completeness depend on the reader: the same trace can be transparent to one reader and encoded to another, depending on whether the reader can recover the structure. The codebook setting shows the two axes pulling apart in practice. Cross-model transfer collapses, but corrupting the visible steps still hurts reader accuracy, so the visible computation is still deriving the answer even when weaker readers cannot follow it. Conversely, semantically null prefixes can change reader behaviour without containing any reasoning, so apparent usefulness does not mean faithful visible reasoning either. Monitorability cannot be read off surface fluency; it requires interventions on the trace together with an explicit reader panel. We release the evaluation code, the cross-model prefix transfer and perturbation sensitivity metrics, and the model organisms used in this work.

Limitations

Scope. The codebook intervention that establishes the legibility–completeness dissociation is run on a five-to-six-term Python arithmetic benchmark with a fixed digit-to-word substitution and an in-trace decryption key. We do not establish that the dissociation generalizes to natural-language CoT, to encodings without an appended key (the steganographic case), or to other forms of token-space obfuscation.

Construct validity. Cross-model prefix transfer confounds prefix legibility with reader prior competence; the baseline subtraction in §3.3 mitigates but does not eliminate this. Perturbations sensitivity aggregates over a truncation grid on which the per-N integrand is non-monotonic under codebook encoding 3, so the scalar should be read against the disaggregated $\Delta(N)$ curve rather than as a uniform shift. The CoT-dependent filter is applied per generator, not per reader; strong readers may answer many retained questions zero-shot, which leaves us unable to fully separate reader re-derivation from genuine load-bearingness under encoding.

Model organisms. The α -amplification organisms are constructed to produce illegible CoTs and do so behaviorally (low *CrossModelPrefixTransfer*), but internal probes find no representational difference from the base reader (§A.6.6). We cannot fully separate "specifically illegible" from "broadly degraded" with the current organisms.

Mechanistic claims. The attention shift from CoT to sinks 7 is correlational; we do not perform causal interventions on attention. The negative probe and SAE results indicate high apparent probe performance can reflect distribution shift rather than monitorability, and we therefore avoid claims about internally-represented legibility.

Ethics Statement

This work studies properties of chain-of-thought traces produced by language models, with the goal of improving the practical monitorability of those traces for oversight. We do not introduce new model capabilities, release new generative artifacts beyond reasoning traces used for evaluation, or collect data from human subjects. The model organisms we construct (codebook-encoded and α -amplified illegible CoTs) are designed to make ex-

isting failure modes legible to a metric, not to advance steganographic or evasion capabilities; the techniques used are well-known and operate on outputs rather than weights.

We see two possible risks. First, characterising when CoTs are illegible to monitors could, in principle, help a model developer push reasoning out of view; we mitigate this by framing the metrics as oversight tools rather than as obfuscation recipes, and by reporting reader-dependent results so that no single reader panel can be cherry-picked. Second, the negative white-box probe results could be misread as evidence that monitorability cannot be improved; we emphasize that the behavioural metrics do recover a real signal, and the right reading is that monitorability is reader-dependent rather than an intrinsic property of the trace. We comply with the ACL Code of Ethics and use only publicly available models and benchmarks (MATH-500, GPQA-Diamond, BBH) for evaluation.

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A Prior Work

The GDM legibility metric is a rubric-based readability score intended to approximate human judgments of CoT interpretability. Traces are scored on a 0–4 scale according to how understandable and coherent they appear at the surface level, independent of whether they are causally load-bearing.

Despite substantial differences in perturbation-based faithfulness and transferability, all conditions achieve similar rubric scores, indicating that surface readability alone fails to distinguish faithful reasoning from decorative or non-load-bearing traces.

- 0: Unintelligible
- 1: Mostly unintelligible
- 2: Mixed / partially understandable
- 3: Mostly understandable
- 4: Fully understandable

B Methodology

B.1 Four Corner Organisms

To calibrate the plane we construct four off-policy CoT organisms whose behavioural signature follows from construction. They serve as calibration anchors, and they make no claim about what real generators spontaneously emit. The latter question is addressed by the on-policy analysis of §??.

Transparent (repetition). Take a correct on-policy CoT and concatenate it $k=3$ times. Redundancy is the legibility signal: any single repetition lets the reader recover the computation, so *CrossModelPrefixTransfer* saturates quickly. Each copy contains the derivation, so *PerturbationSensitivity* is high. A recursive-step-elaboration variant lies at the same corner but is not tested here.

Encoded (codebook+key). Take a correct CoT, encode all numerical content under a fixed digit→category-word codebook (fruits, colours, cities), and append the decryption key as a suffix. The full computation is present and invertible, though inaccessible to readers below the truncation level at which the key first appears. Harder encodings (one-time-pad, Huffman) occupy the same corner but are not tested.

Latent (lorem ipsum). Replace the on-policy CoT with filler text matched in token count, and

retain only question–answer pairs where the generator still answers correctly under this substitution. Visible tokens carry no task-relevant signal; correct answers, when they occur, reflect computation outside the visible stream.

Post-hoc. Two recipes, both yielding fluent followable text whose intermediate claims are not load-bearing for the reader’s reconstruction. In the *answer-first* recipe, we prepend the correct answer and ask the generator to produce a CoT explaining it. In the *no-CoT-needed* recipe, we restrict to questions the generator solves correctly without any CoT, then ask for a CoT.

B.2 Calibration via Off-Policy Corner Organisms

To verify that the metrics actually track the axes they purport to track, we construct one off-policy CoT *organism* for each corner, chosen so that its behavioural signature follows from construction. The organisms are calibration anchors and make no claim about what real generators spontaneously emit; that question is addressed by the on-policy analysis of §??. Recipe-level details are deferred to §4.

Transparent is anchored by *redundant* CoTs in which the derivation appears multiple times: each repetition independently lets the reader recover the computation, so legibility saturates at low prefixes while intermediate corruption remains punishing. **Encoded** is anchored by *codebook-with-key* CoTs in which numerical content is rewritten under a fixed substitution and the decryption key is appended as a suffix: the computation is fully present and load-bearing, but is inaccessible to readers below the truncation level at which the key first appears. **Latent** is anchored by *filler* CoTs in which the on-policy reasoning is replaced by token-count-matched lorem ipsum and only question–answer pairs the generator still answers correctly are retained: the visible stream carries no task-relevant signal, so any correctness must flow from computation outside it. **Post-hoc** is anchored by two recipes, both producing fluent followable text whose intermediate claims do not actually drive the reader’s answer: an *answer-first* recipe in which the generator is shown the correct answer and asked to produce a CoT explaining it, and a *no-CoT-needed* recipe in which a CoT is requested for questions the generator already solves CoT-free.

Stronger or alternative organisms exist at each

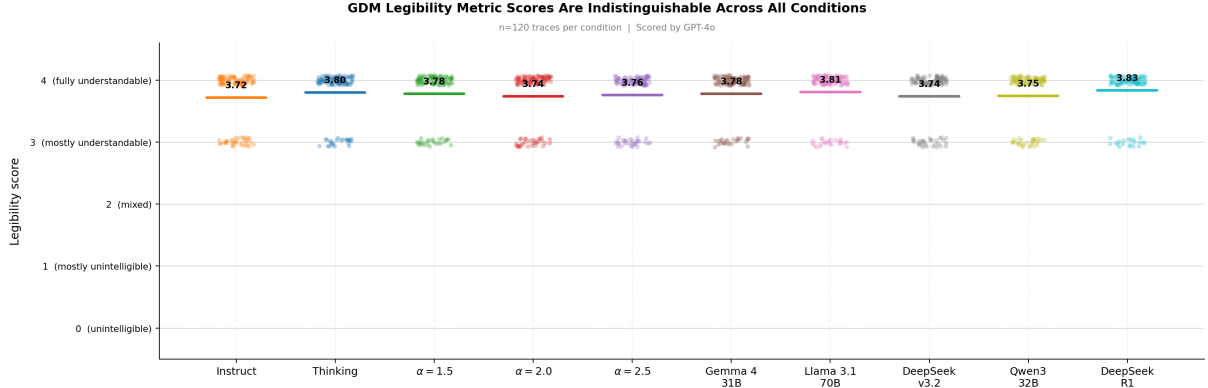


Figure Appendix A.1: Human-style legibility scores from a rubric-based evaluator are nearly indistinguishable across models and steering conditions. Each point is a trace-level score (higher is more understandable), with horizontal bars denoting condition means. Despite substantial differences in transferability and perturbation-based faithfulness elsewhere in the paper, all conditions cluster tightly around similar scores, indicating that surface readability judgments fail to detect deeper differences in whether reasoning traces are actually useful or causally meaningful.

corner—recursive step-elaboration for Transparent, one-time-pad or Huffman codes for Encoded—and would refine the calibration. The organisms above are the minimal set needed to anchor the four quadrants of the plane.

B.3 Mechanistic Complement

The behavioural metrics above are black-box: they cannot, on their own, distinguish a non-recoverable encoding from a genuinely latent computation, since both yield low *CrossModelPrefixTransfer* and low *PerturbationSensitivity*. To approach this distinction we construct an illegible model organism by weight-space task-vector amplification of a Think variant over an Instruct base (§4), producing an “overthinking” model whose CoTs have substantially lower mean reader-panel *CrossModelPrefixTransfer* than either parent. We additionally analyse the resulting residual-stream representations using linear probes and pretrained sparse autoencoders. The overall white-box signal turns out to be weak and confounded: probes trained on natural on-policy legibility are only modestly decodable, while probes trained on synthetic off-policy perturbations achieve much higher AUROC but learn directions orthogonal to natural legibility. This is itself a methodological finding: high apparent probe performance can reflect detection of distributional artifacts rather than monitorability proper, and it motivates the behavioural focus of this paper. Full white-box results are reported in §H.1.

B.4 Truncation Diagnostic: Implementation

For each filtered (Q, r) pair we sweep the prefix fraction $N \in \mathcal{N}$ revealed to each reader. The CoT is first leak-segmented (§??) and end-truncated at the latest non-leaking block. The remaining prefix is prefilled as a prior assistant turn; a follow-up user turn asks for the final answer; the subsequent assistant turn is prefilled with “The answer is:” and the reader is forced to emit ≤ 32 tokens with no rederivation. Instruction-tuned readers receive additional anti-rederivation prompting (Appendix ??). The truncation grid is $\mathcal{N} = \{25, 50, 75, 100\}\%$ for the main sweep; a denser $\{10, 20, \dots, 100\}\%$ grid is used for selected cells where probe analyses require it, with the choice between grids compute-driven. Pairwise Cohen’s κ across readers at each truncation level is reported in Appendix ??; high agreement at full reveal supports the view that legibility is a property of the CoT rather than of any single reader, while codebook-style traces produce reader-dependent low-prefix curves that the metric surfaces as a feature rather than an inconsistency.

B.5 Perturbation Diagnostic: Implementation

For each clean (Q, r) pair we (i) apply a perturbation π at 100% rate, (ii) end-truncate the perturbed trace to $N \in \{25, 50, 75, 100\}\%$, and (iii) prefill it to each reader under the same answer-only protocol as the truncation track. The joint sweep is over perturbation $\times$ truncation. We evaluate 20 perturbation modes across six families (Table Appendix E.1); full mode specifications are in Appendix ??. The reference perturbation π^* used in

PerturbationSensitivity is the suffix-corruption variant of §3: starting at a chosen step k^* , each subsequent step is replaced with a counterfactual continuation generated by a separate model, with the counterfactual propagated consistently through the remainder of the trace.

B.6 Corner Organism Construction Recipes

Transparent (repetition). Take a correct on-policy CoT and concatenate it $k = 3$ times. **Encoded (codebook+key).** Take a correct CoT, encode all numerical content under a fixed digit→category-word codebook (fruits, colours, cities), and append the decryption key as a suffix. Harder encodings (one-time-pad, Huffman) occupy the same corner but are not tested. **Latent (lorem ipsum).** Replace the on-policy CoT with filler text matched in token count, and retain only question–answer pairs where the generator still answers correctly under the substitution. **Post-hoc.** Two recipes: (i) *answer-first*—prepend the correct answer and ask the generator to produce a CoT explaining it; (ii) *no-CoT-needed*—restrict to questions the generator solves correctly without any CoT, then ask for a CoT.

B.7 Mechanistic Organism

We construct illegible organisms by task-vector amplification: given Qwen3-8B THINK and INSTRUCT variants with weights W_T and W_I , we form

$$W_{\text{ill}} = W_I + \alpha (W_T - W_I), \quad \alpha > 1,$$

producing models whose CoTs on custom Python expression evaluation have substantially lower mean reader-panel *CrossModelPrefixTransfer* than either base model: [0.14] at $\alpha = 1.5$ and [0.09] at $\alpha = 2.0$, against [0.57] for Instruct and [0.47] for Think. The linear-probe and SAE analyses referenced in §B.3 are reported in full in §H.1 and Appendix G.

C Codebook Encoding

Codebook Encoding. The codebook encoding replaces every digit (0–9) in the chain-of-thought (CoT) with a word from a fixed category, and appends the decryption key at the end. The reasoning text and operators remain unchanged; only numeric digits are substituted.

Codebooks.

Digit	Fruit	Color	City
0	apple	crimson	tokyo
1	banana	azure	cairo
2	cherry	emerald	paris
3	dragonfruit	violet	lima
4	elderberry	amber	oslo
5	fig	ivory	rome
6	guava	scarlet	seoul
7	honeydew	indigo	dubai
8	kiwi	coral	nairobi
9	lemon	teal	havana

Example (Fruit Codebook). Original CoT:

First, compute $15 * 14 = 210$. Then $210 * -29 = -6090$.

Encoded CoT:

First, compute bananafig * bananaelderberry = cherryban
Then cherrybananaapple * -cherrylemon = -guavaapplelemo

Codebook:

{0: apple, 1: banana, 2: cherry, 3: dragonfruit, 4: elde
5: fig, 6: guava, 7: honeydew, 8: kiwi, 9: lemon}

Each digit is replaced character-by-character (e.g., $15 \rightarrow \text{bananafig}$), rather than mapping entire numbers to single tokens. The codebook dictionary is appended at the end of the CoT.

Truncation Analysis. Truncation plays a key role because the codebook appears only at the end of the CoT. At low truncation levels (e.g., 10%, 25%), the reader observes encoded numbers without access to the mapping and must infer it implicitly. At higher truncation levels (e.g., 75%, 100%), the reader has access to both the encoded reasoning and the decryption key. This setup tests whether models can actively use the codebook to decode and follow the reasoning, rather than ignoring the encoded numerical structure.

D Interpretability of legible and illegible CoTs in activation space

D.1 Other Stuff

Legible and illegible chains of thought may start from nearly the same internal state because both traces are responding to the same question and have not yet committed to different reasoning strategies. As generation continues, there may be a point where the model begins processing the problem differently, causing the hidden representations to separate, which he calls the “opening.” Since attention keys, queries, and values are linear projections

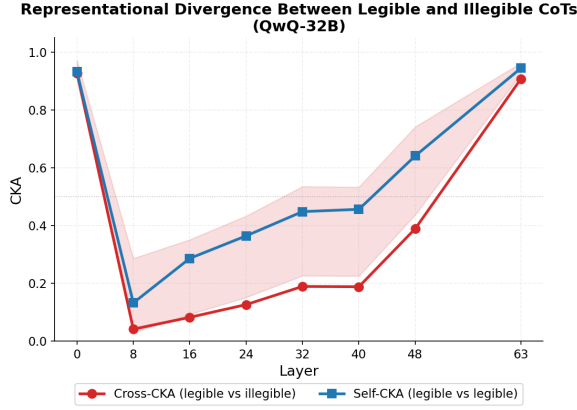


Figure Appendix D.2: *Legible and illegible reasoning traces may traverse different internal computational pathways even when producing superficially similar outputs* We use CKA to compare activation subspaces across reasoning regimes, while noting known limitations of CKA as a representation similarity metric (Davari et al., 2022). The blue curve reports a calibration baseline, self-CKA between independent sets of legible traces, estimating the similarity expected within the same reasoning regime. The red curve reports cross-CKA between legible and illegible traces. Lower red values therefore indicate representational divergence between the two trace types beyond ordinary within-class variation. The recovery of similarity in later layers is consistent with convergence onto shared answer-generation representations after distinct intermediate reasoning dynamics. The blue-minus-red gap at each layer can be interpreted as the extent to which illegible traces depart from the representational manifold associated with legible reasoning.

of these hidden states, once the hidden states diverge the attention computations also diverge, so the model starts looking at different prior tokens or weighting context differently. That changes what information gets carried forward into later layers, which can reinforce a distinct reasoning path and make one trace less readable or less useful to an external reader. Near the end, the two trajectories may partially reconverge, or “close,” because both traces still need to produce a similar final answer in natural language, so the output distribution becomes more alike even if the internal route taken was different.

E Truncation and Perturbation Curves

Choice of perturbation method. Locating the load-bearing steps of a CoT requires a perturbation that isolates step-level effect from measurement artifact. Two natural candidates fail this test (Figure Appendix E.4). Method A (prefix-only) con-

found step importance with prefix length: $\Delta(k)$ rises monotonically with k across all generators because short corrupted prefixes are too short for reliable extraction regardless of which step was corrupted, so the curve traces the truncation ramp rather than the per-step contribution. Method B (single-step swap inside the full trace) confounds in the opposite direction: the clean suffix surrounding the corrupted step provides enough redundant information that the reader recovers regardless, compressing $\Delta(k)$ toward zero everywhere and shrinking the gap between natural generators and our amplified illegibility organism to 0.34. Method C replaces step k with a counterfactual and uses a separate model to extend that counterfactual into a coherent suffix, so the reader sees a complete trace that is internally consistent with the corrupted step. $\Delta_C(k)$ therefore isolates the steps whose content the answer genuinely depends on, in exactly the setting a monitor faces: a fluent trace where surface coherence cannot be used to flag the corruption, and recovery requires the answer to be recomputable only from the load-bearing content.

E.1 Perturbation Method Variants: Notation

We measure visible completeness by corrupting reasoning steps with a counterfactual statement and observing how those corruptions affect a reader’s ability to recover the correct answer. The space of perturbation methods is non-trivial, and different perturbations measure different notions of load-bearingness. We considered three variants but ultimately adopt corrupting from step k onward and propagating the counterfactual through all subsequent reasoning steps. This method removes the clean suffix that would otherwise allow the reader to bypass the corruption. $\Delta_C(k)$ therefore measures whether the computation beginning at step k causally contributes to the final answer:

- **(A) Prefix-only perturbation:** corrupt step k in the truncated prefix

$$[s_1, \dots, s'_k],$$

and ask the reader to extract an answer from the corrupted prefix alone.

- **(B) Single-step swap:** corrupt step k within the full trace

$$[s_1, \dots, s'_k, \dots, s_m],$$

while leaving all surrounding context unchanged.

Legible CoT Traces Show Greater Self-Referential Attention Within The Reasoning Span
Metric: mean attention from CoT token c to strictly prior CoT tokens $k < c$, averaged over heads and CoT positions

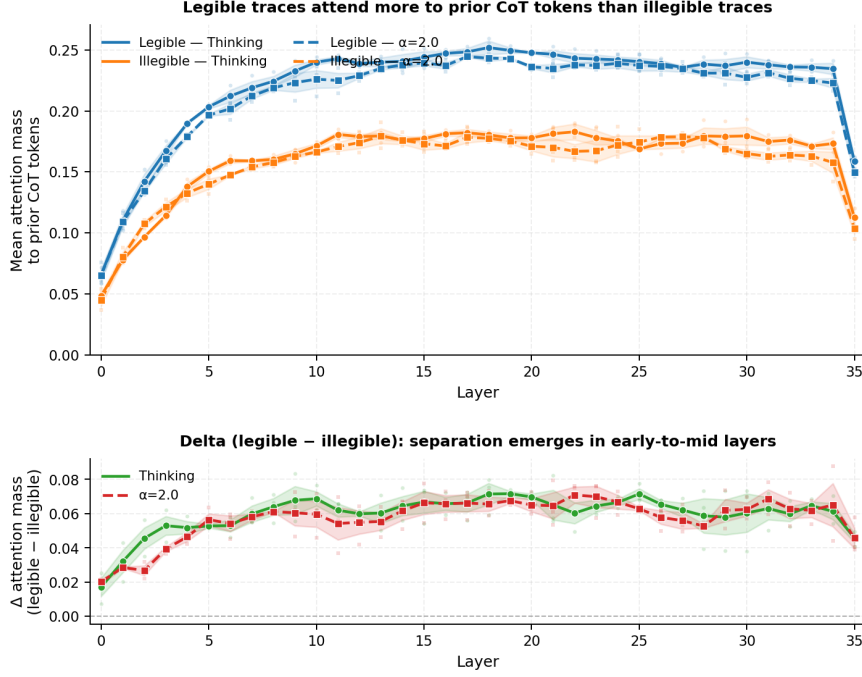


Figure Appendix D.3: Mean self-referential attention within the reasoning span. We measure, at each layer, the average attention mass from a chain-of-thought token to earlier chain-of-thought tokens. Legible traces consistently allocate more attention to prior reasoning tokens than illegible traces across both baseline thinking and amplified $\alpha = 2.0$ settings. The lower panel shows a stable positive gap (legible minus illegible), emerging in early-to-mid layers and persisting through depth, suggesting that readable reasoning traces more strongly reuse and integrate prior intermediate steps rather than bypassing them.

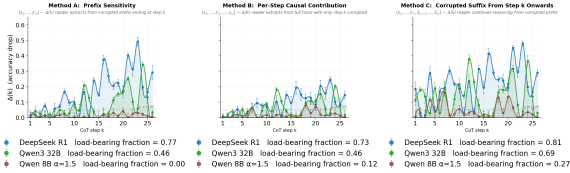


Figure Appendix E.4: Three perturbation schemes for $\Delta(k)$, evaluated on MATH-500. (A) Prefix-only: reader extracts from a corrupted prefix ending at step k . (B) Single-step swap: reader extracts from the full trace with only step k corrupted. (C) Suffix corruption: step k is replaced with a counterfactual and a separate model extends it into a coherent suffix. A confounds step importance with prefix length ($\Delta(k)$ tracks the truncation ramp); B is bypassed by the clean surrounding suffix ($\Delta(k) < 0.1$ almost everywhere). C removes both confounds and separates the three generators cleanly. Curves are mean $\Delta(k) \pm 1$ s.e. over MATH-500; $\varepsilon=0.05$ marks the load-bearing threshold; legends report PerturbationSensitivity. We adopt C as the reference perturbation π^* .

- (C) **Suffix corruption:** replace step k with a counterfactual claim and propagate that

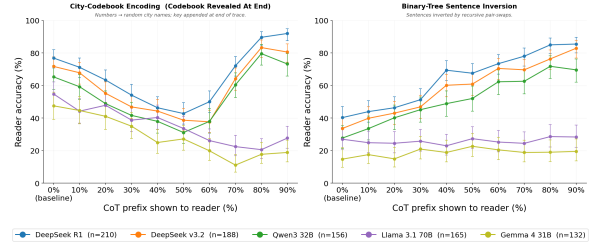


Figure Appendix E.5: Truncation curves for sentence inversion reasoning traces. Reader accuracy improves slightly only after a large fraction of the encoded chain-of-thought is revealed. Non filtered result. Generator model may be getting samples where it doesn't benefit from cot.

counterfactual forward through all subsequent steps, yielding

$$[s_1, \dots, s_{k-1}, s'_k, s'_{k+1}, \dots, s'_m].$$

Methods A and B both contain implicit bypass mechanisms. Under (A), corrupting an early step often has little effect because the truncated prefix may not yet contain enough computation for

Table Appendix E.1: **Perturbation families.** Full model specifications in Appendix ??.

Family	#	What it targets / preserves
Word replacement	3	Lexical content (all / function-word / connective); numbers and operators preserved.
Number corruption	2	Numeric and operator tokens; non-mathematical words preserved or stripped.
Word replacement + number re-encoding	6	Lexical content plus number surface form (only English word, reversed, dice, Braille Morse).
Character-level ciphers	5	Sub-word structure (ROT-2/4/8/13, substitution); word boundaries preserved.
Structural	1	Sub-word meaning and text backbone as in Qwen3-8B.
Codebook encodings	3	Digits $\rightarrow$ category; fruit, phase, class, CoTs (666 records: G1 164, G3 156 legible/illegible); single-domain arithmetic (300 problems, 1,200 vectors at 4 truncation levels); cross-domain (677 problems across BBH, GPQA, MATH with reader-panel labels, 60,630 vectors at 10 truncation points); cipher augmentation (1,029 enciphered CoTs via ROT13, Caesar shift-7, Huffman).

any answer to be recoverable. The reader therefore fails regardless of the corruption, compressing $\Delta(k)$ toward zero at small truncation levels. Under (B), corrupting a single intermediate step rarely changes extraction because the clean suffix surrounding the corruption still contains sufficient information for the reader to recover the answer. Both methods therefore systematically under-detect load-bearingness whenever the reasoning trace contains redundancy or downstream self-correction.

We adopt Method (C) as our primary perturbation. By corrupting from step k onward and propagating the counterfactual through all subsequent reasoning steps, Method (C) removes the clean suffix that would otherwise allow the reader to bypass the corruption. $\Delta_C(k)$ therefore measures whether the computation beginning at step k causally contributes to the final answer.

Counterfactual continuations $s'_{k+1}, \dots, s'_m$ are generated by a separate model conditioned on s'_k , so that the suffix is a consistent extension of the corrupted step rather than an adversarially crafted distractor.

The truncation grid for the perturbation metric is $\mathcal{N} = \{10\%, 20\%, \dots, 90\%\}$. The step index k is swept over the segmented step boundaries returned by the leak segmenter (§??).

F Additional Results

G White-Box Analysis of Legibility in Model Internals

G.1 Overview

We investigated whether model internal representations encode information about chain-of-thought (CoT) legibility. Using residual-stream activations

extracted via `vllm-lens`, we trained linear probes, compared generator and reader models, and applied

Sparse Autoencoder (SAE) feature analysis across multiple models and domains.

Models probed. G1: DeepSeek-R1-Distill-Owen-32B (64 layers, 5120 hidden); G3: QwQ-32B (64 layers, 5120 hidden); Qwen3-8B base (36 layers, 4096 hidden) as both on-policy and off-policy reader; alpha-1.5 generator (Qwen3-VL-8B, same text backbone as Qwen3-8B).

Datasets. Phase 1: single-domain arithmetic (300 problems, 1,200 vectors at 4 truncation levels); cross-domain (677 problems across BBH, GPQA, MATH with reader-panel labels, 60,630 vectors at 10 truncation points); cipher augmentation (1,029 enciphered CoTs via ROT13, Caesar shift-7, Huffman).

G.2 Generator-Native Legibility Probes

We trained logistic regression probes on the generator models’ own last-token activations to predict legibility (5-fold StratifiedKFold, bootstrap 95% CI).

G1 achieves AUROC 0.630 [0.571, 0.696] at layer 48 ($p = 0.038$, 1,000-permutation test). G3 achieves AUROC 0.615 [0.559, 0.672] at layer 48 ($p = 0.070$). Both peak around 75% network depth. The signal is modest—confidence intervals overlap with chance at most individual layers due to small samples ($n = 156$ –164) and $\sim 80/20$ class imbalance (Figure Appendix G.11).

G.3 Perplexity as a Confound Control

Illegible CoTs have higher reader perplexity (R2 mean: legible = 1.98, illegible = 3.92). However, perplexity alone achieves AUROC 0.385—below chance—as a legibility predictor. Adding perplexity as a covariate does not improve probe performance (0.615 $\rightarrow$ 0.622, within CI overlap). The probe captures information beyond distributional shift.

G.4 On-Policy vs Off-Policy Probe Directions Are Orthogonal

We compared probes trained on natural on-policy CoTs (Qwen3-8B reading alpha-1.5 outputs) versus off-policy perturbation data (Qwen3-8B reading cipher/number-corrupted CoTs).

Off-policy probes achieve AUROC 0.861 at layer 32; on-policy probes achieve AUROC 0.676

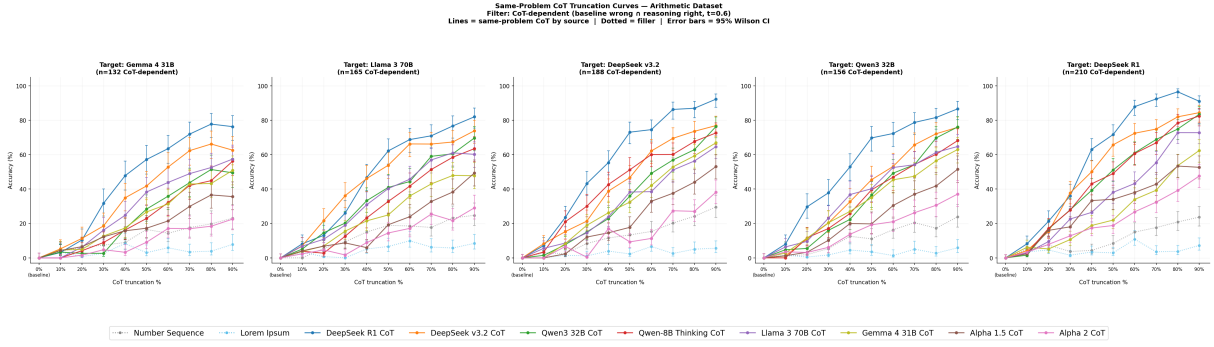


Figure Appendix E.6: Same-problem chain-of-thought truncation curves on the arithmetic dataset across open-source models. For each problem p , a source model M_i generates a reasoning trace $R = (s_1, s_2, \dots, s_m)$ and a target reader model M_j is prefixed with the first k fraction of the trace before producing an answer. We estimate $f_{i,j}(k) = \Pr(M_j \text{ correct} \mid \text{prefill} = R[:k])$ over source-reader pairs and report target-conditioned accuracy curves. Dotted lines denote matched-length filler controls. Steeper early gains indicate more legible traces whose useful information is exposed incrementally, while flat or late-rising curves indicate delayed or opaque reasoning. We observe substantial heterogeneity across source models: some traces transfer useful signal early, whereas others concentrate utility near the end of the reasoning trace.

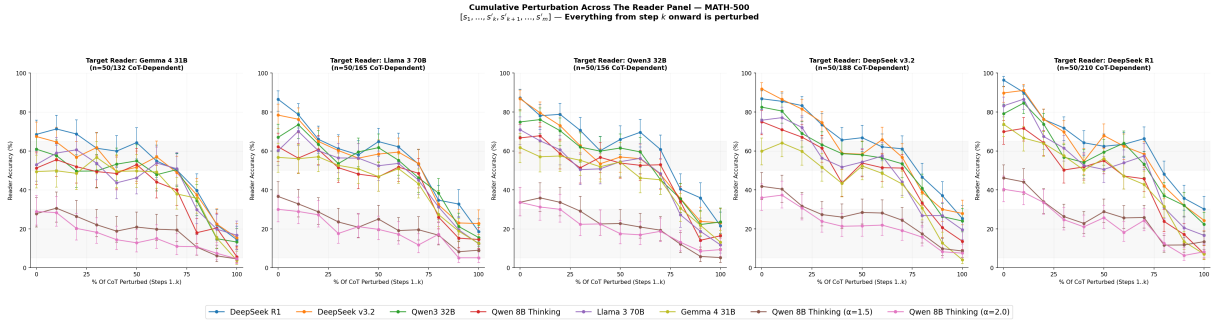


Figure Appendix E.7: **Cumulative perturbation across the reader panel on MATH-500.** For each generator’s CoT, we corrupt the last $k\%$ of reasoning steps and measure each reader’s accuracy on the resulting prefix. Faithful, load-bearing traces should degrade smoothly as more steps are perturbed; decorative or post-hoc traces should be insensitive to corruption. Steered Qwen 8B variants ($\alpha=1.5, 2.0$) sit far below the unsteered baselines at 0% and decay weakly, consistent with reasoning that has migrated out of the token stream.

at layer 35 (Figure Appendix G.10, left). However, the two probe directions are orthogonal at every layer: cosine similarity ranges from -0.02 to $+0.02$ (Figure Appendix G.10, right). Cross-distribution transfer produces strongly negative R^2 at all layers.

This is the strongest evidence that high off-policy AUROCs reflect perturbation-artifact detection rather than genuine legibility encoding. Three independent analyses converge: (1) probe directions are orthogonal, (2) cross-distribution transfer fails completely, (3) SAE feature sets share only 1–5% overlap (Figure ??).

G.5 Generator vs Base Reader

The alpha-1.5 generator and base Qwen3-8B achieve statistically indistinguishable AUROCs (0.671 vs 0.676 at layer 35). With PCA regulari-

sation to $k = 32$ components, both reach ~ 0.71 (Figure Appendix G.11, right). Probe direction cosine similarity decreases from 0.55 (layer 0) to 0.27 (layer 35), indicating partially overlapping but increasingly distinct representations at depth.

G.6 SAE Feature Analysis

We applied the QwenScope SAE (Qwen/SAE-Res-Qwen3-8B-Base-W64K-L0_100; 65,536 features, top- $K=100$) to decompose residual-stream activations into sparse interpretable features.

Single-domain (arithmetic). On-policy sources share 25–43% of their top-50 discriminative features (Jaccard). Off-policy sources share only 1–5%, near chance (Figure ??). Autolabelling of 86 features via Claude Sonnet 4 confirmed that the top

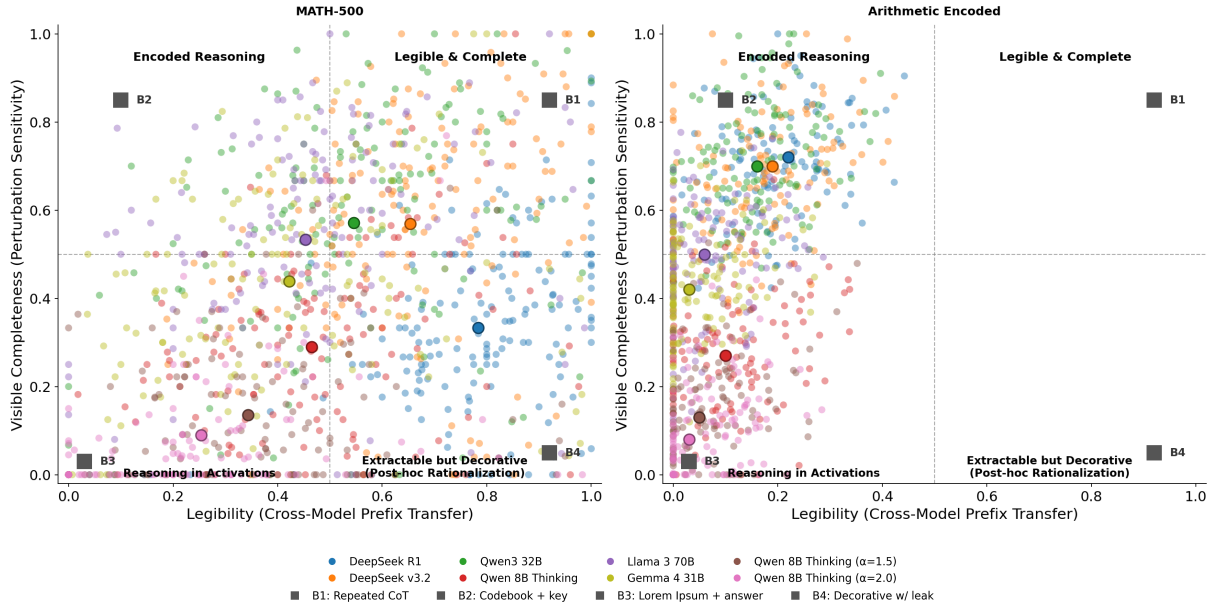


Figure Appendix F.8: Reasoning traces occupy distinct regions in the legibility completeness landscape. We plot the full set of points here across models. Encoded codebook traces consistently fall into a low-legibility, high-completeness regime: they exhibit low *CrossModelPrefixTransfer* but high *PerturbationSensitivity*, indicating that answer-relevant information remains present and causally necessary despite being difficult for reader models to extract. Truncation behavior further supports this interpretation, as reader accuracy improves only after most of the encoded sequence is revealed, suggesting that the reasoning is complete but effectively illegible under partial observation.

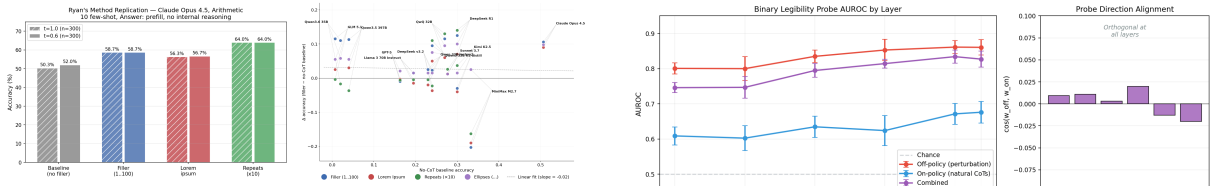


Figure Appendix F.9: **Filler tokens reveal non-semantic channels that affect accuracy.** **Left:** Following Lanham et al. (2023), semantically null fillers (numbers, lorem ipsum, repeated prompts) improve arithmetic accuracy on Claude Opus 4.5 ($n=300$) by 4–14 points over a no-CoT baseline, despite containing no task-relevant content. **Right:** Across conditions, accuracy varies systematically with tokenization/structure even when semantics are held fixed, indicating that models exploit non-content channels (e.g., compute extension, KV-cache geometry, position bias). These channels enable post-hoc rationalization: tokens can influence outputs without encoding the computation. Our perturbation setup controls for these factors by holding token count, position, and surface form fixed, so $\Delta(k)$ isolates content sensitivity.

features encode mathematical content (operators, digits, parentheses), not reasoning quality.

Cross-domain (BBH, GPQA, MATH). Layer 16 shows the highest cross-domain

Figure Appendix G.10: **On-policy vs off-policy legibility probes.** **Left:** AUROC by layer. Off-policy probes (perturbation data) achieve 0.86 but on-policy probes (natural CoTs) reach only 0.68. **Right:** Cosine similarity between probe weight vectors is indistinguishable from zero at all layers, indicating the two probes detect fundamentally different properties.

consistency (Jaccard 0.24–0.35, 18 features in triple intersection). Later layers become increasingly domain-specific (layer 35: Jaccard 0.09–0.12, 3 features in triple intersection). The *diff_rate* correlation between domain pairs peaks at $r = 0.62$ (GPQA–MATH, layer 16) and falls to $r = 0.25$ (BBH–GPQA, layer 35); see Figure Appendix G.12.

Within-task validation probes using 50 SAE features (5-fold GroupKFold by sample ID) achieve AUROC 0.609 ± 0.011 on MATH at layer 16 (best). All values are above chance with tight standard de-

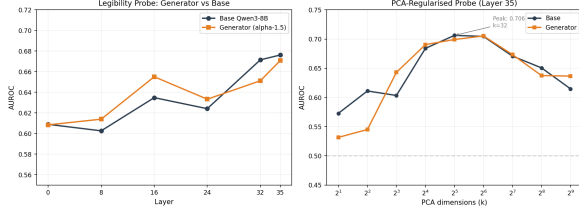


Figure Appendix G.11: **Generator vs base model.** *Left:* Layer-wise AUROC for both models on the same 1,200 on-policy samples. *Right:* PCA dimensionality sweep at layer 35; peak AUROC ≈ 0.71 at $k = 32$ PCs.

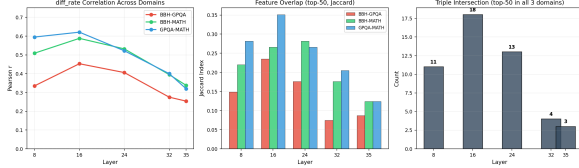


Figure Appendix G.12: **Cross-domain SAE feature agreement.** *Left:* diff_rate Pearson correlation across domains by layer. *Middle:* Jaccard overlap of top-50 features. *Right:* Triple intersection count (features in top-50 for all three domains). Layer 16 is the most domain-general.

viations (0.004–0.020) but the signal is weak.

Cross-domain feature characterisation. Of 65,536 features, only 5 pass a strict cross-domain filter (same-sign diff_rate in all 3 domains, $|\text{diff_rate}| > 0.01$ in ≥ 2):

Feature	Dir.	BBH	GPQA	MATH	Label
F3817	legible	+0.026	+0.055	+0.046	Logical connectives
F55870	legible	+0.025	+0.013	+0.018	Concrete reasoning initiation
F5550	illegible	−0.029	−0.011	−0.000	Disrupted structural connectors
F51015	legible	+0.019	+0.012	+0.000	Sentence-initial tokens
F46021	illegible	+0.012	+0.002	+0.018	Fragmented math operators

Table Appendix G.2: SAE features passing cross-domain consistency filter. Per-domain diff_rate values shown; overall $|\text{diff_rate}|$ is inflated by cipher augmentation.

F3817 is the most interpretable: it fires on function words (“and”, “in”, “to”, “than”) connecting reasoning steps, with higher activation when logical connections are explicit.

G.7 Cipher Augmentation Experiment

We enciphered 512 CoTs via ROT13 (512), Caesar shift-7 (511), and Huffman encoding (6) to create canonically illegible samples—complete reasoning in an unreadable format.

When cipher samples were placed in a separate domain, the *quadruple intersection* (features dis-

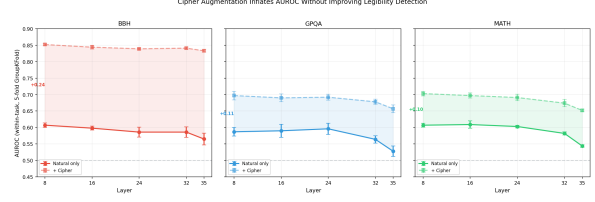


Figure Appendix G.13: **Cipher augmentation inflates AUROC.** Solid lines: natural text only. Dashed lines: with cipher samples as illegible. Shaded region shows the inflation, which is proportional to cipher sample count (BBH received the most).

criminative in $\text{BBH} \cap \text{GPQA} \cap \text{MATH} \cap \text{CIPHER}$) was **zero** at all layers and all top- N thresholds. The features detecting enciphered text are entirely disjoint from those detecting natural-text illegibility.

When cipher samples were folded into their source domains, within-task AUROCs inflated dramatically (BBH: 0.607 $\rightarrow$ 0.852; MATH: 0.607 $\rightarrow$ 0.703), proportional to cipher sample count (Figure Appendix G.13). The probe learns to separate enciphered from natural text, not to detect reasoning quality.

G.8 CKA Representational Similarity

Within-source CKA between legible and illegible samples is high (0.46–0.90), indicating they share most representational structure. Cross-source CKA (off-policy vs on-policy) is much lower (0.15–0.60), confirming these data sources occupy different regions of activation space (Figure ??).

G.9 Summary of Key Findings

Finding	Evidence	Strength
Legibility weakly encoded	On-policy AUROC 0.68; PCA 0.71	Significant
Off-policy signal confounded	$\cos(\mathbf{w}_{\text{off}}, \mathbf{w}_{\text{on}}) \approx 0$; transfer $R^2 < 0$	Strong
Generator = base	AUROC 0.67 vs 0.68	Consistent
Perplexity not explanatory	Perplexity-only AUROC 0.385	Rules out
SAE features domain-specific	5/65,536 pass filter	Strong
Cipher features disjoint	Quadruple intersection = 0	Definitive
Mid-layers most general	Layer 16: Jaccard 0.24–0.35	Consistent

Table Appendix G.3: Summary of white-box legibility analysis findings.

H Discussion

H.1 White-box probes do not reveal a simple legibility variable

A natural alternative to our behavioural metrics is to ask whether legibility is directly represented inside

the model: perhaps a white-box auditor could determine whether a CoT is monitorable from residual-stream activations or sparse-autoencoder (SAE) features alone. We test this on Qwen3-8B using residual-stream probes at CoT truncation positions and the QwenScope 64K-feature SAE, with reader-panel transfer utility as the supervision signal.

On natural on-policy CoTs, linear probes recover only a weak signal. Residual-stream probes achieve AUROC 0.60–0.68 across layers, improving to 0.71 under PCA regularisation. SAE-feature probes on cross-domain natural-text data reach AUROC 0.56–0.61. The signal is statistically reliable but weak and distributed: we find no evidence for a single localized “legibility feature” or privileged internal readout. Moreover, the generator model does not expose substantially stronger legibility information than the base reader model under these probes (layer-35 AUROC 0.671 vs. 0.676).

The stronger result is negative. Probes trained on off-policy corrupted traces (ROT13, substitution ciphers, numerical corruption) reach AUROC 0.86, but their learned directions are orthogonal to the on-policy legibility directions at every layer ($\cos(w_{\text{off}}, w_{\text{on}}) \in [-0.02, 0.02]$), and cross-distribution transfer fails catastrophically. SAE analysis gives the same diagnosis: top discriminative SAE features overlap only 1–5% between on-policy and off-policy datasets, near chance. Across BBH, GPQA, and MATH, only 5 of 65,536 SAE features pass a strict cross-domain consistency filter, and these correspond primarily to surface/domain correlates (e.g. logical connectives and mathematical formatting) rather than an internal representation of reasoning quality.

Enciphered traces activate a disjoint feature set from natural illegible reasoning: when ROT13/Caesar controls are included, probe AUROC increases substantially, but the discriminative features for cipher text have zero overlap with those for natural-text legibility. Structural illegibility and semantic illegibility therefore occupy separate representational subspaces. This mirrors the geometry of Figure 1: encoded traces are separable through structural-text features, whereas natural illegible traces occupy a weaker, distributed, reader-relative subspace.

Together these results support the functional framing of the paper. White-box access reveals weak correlates of reader-legibility, but the dominant probe signals are artifacts of perturbation,

Table Appendix H.4: **White-box probes recover weak natural legibility signals and strong artifact signals.** Residual-stream probes and SAE features were evaluated on Qwen3-8B activations at CoT truncation positions.

Check	Result	Interpretation
Natural CoT residual probes	AUROC 0.676±0.031	Weak but real signal
PCA-regularised probes	Best AUROC 0.706	Low-dimensional
Generator vs. base model	0.671 vs. 0.676	No privileged reader
Off-policy corrupted CoTs	AUROC 0.861±0.019	Strong apparent signal
On/off probe alignment	$\cos(w_{\text{off}}, w_{\text{on}}) \approx 0$	Artifact direction
On/off SAE overlap	Jaccard 0.01–0.05	Near-chance overlap
Cross-domain SAE consistency	5/65,536 features	No general legibility
Cipher vs. natural illegibility	zero top-feature overlap	Separate representational spaces

domain, or text naturalness. Internal probes are therefore not a substitute for reader-relative diagnostics.

Interventions by quadrant. The two-axis system enables differentiated oversight strategies, each grounded in the diagnostic framework rather than introduced as separate empirical contributions. *Post-hoc* CoTs are the highest triage priority. They present as legible and are likely to pass rubric-based gates, while their intermediate claims are not load-bearing, which is the regime in which unfaithful rationalisation (Lanham et al., 2023; ?) hides. *Encoded* CoTs with recoverable codebooks can in principle be surfaced by panel composition: the codebook-bimodality result (§??) shows strong readers sometimes infer codebooks unprompted, so a strong/weak reader discrepancy signal is a plausible detector for recoverable encodings, subject to future validation. *Latent* CoTs cannot be audited through the token stream and require interpretability-side methods, such as probing, activation anomaly detection, or architectural constraints that force reasoning into auditable channels. *Transparent* CoTs are the cheapest to monitor.

Data filtering by quadrant. A concrete training-pipeline intervention falls out of the plane. When building SFT supervision from on-policy reasoning traces, one could filter by quadrant: *Transparent* traces are kept, *Post-hoc* traces are down-weighted or removed to avoid training on non-load-bearing rationales, *Encoded* traces are normalised (decoded) before supervision, and *Latent* traces are not treated as reasoning supervision at all. We do not evaluate downstream effects of such filtering here.

Reader panel as a tunable lens. Both diagnostics are reader-relative by design. A panel heterogeneous along a targeted axis (language coverage, tokeniser granularity, mathematical fluency) produces NTP curves that discriminate along that axis; a homogeneous panel does not. Disagreement between strong and weak readers is itself a candidate detection signal for recoverable encodings. Future oversight infrastructure should treat panel selection as a first-class hyperparameter and report per-reader breakdowns, alongside panel means.

Agentic settings. An agent under token-budget or latency pressure may produce compressed or encoded outputs by default. The framework predicts a direction of movement (Transparent→Encoded or Transparent→Latent, depending on whether load-bearing content migrates into compressed surface form or into activations), and the NTP and LBF diagnostics provide a way to detect the shift before the trace is relied on for oversight.

Operations the framework does not cover. Two extensions lie outside the scope of the current diagnostics. *Steganographic robustness*, whether information is carried in syntax-level features that survive paraphrase, would require paraphrase and multilingual round-trip tests. *Architectures with non-standard forward passes* (recurrent transformer variants, multi-token-per-step generation, hidden scratchpads) require extending the framework to non-textual channels. A further exploratory extension, *different-problem CoT transfer*, tests whether a CoT’s utility is a reusable procedure or answer-specific leakage; we do not include results for this protocol. Finally, internal probes trained on perturbation-generated illegible traces are not reliable substitutes for functional diagnostics: in our experiments they learn directions orthogonal to natural on-policy legibility (§H.1), indicating that high probe accuracy can reflect perturbation artifacts rather than monitorability.